MOULOUD SERIR

Full-Stack and AI Developer | React, Next.js, TypeScript, Node.js, Python

Sherbrooke, QC (Montréal hybrid or remote) • mouloudserir24@gmail.com

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SUMMARY

Bilingual web developer and CS student graduating in Dec 2026, with two AI and full-stack internships (Intelligence Industrielle) and one web development internship (Borealis). I build web apps end to end with React, Next.js, TypeScript and Node.js and AI features (agents, RAG, tool calls), using AI coding tools daily with TDD, integration and E2E tests. Available from January 2027.

PROFESSIONAL EXPERIENCE

Intelligence Industrielle — Montréal, QC

Software Engineering Intern (second internship) — May 2026 – Aug 2026

TypeScript · React · Next.js · TanStack Query · Node.js · MongoDB · BigQuery · n8n · Playwright

- Built an internal app where **AI agents orchestrated in n8n** create versioned BI reports from live BigQuery data, edited via chat.
- **Redesigned the UI/UX** of an internal project lifecycle app with my supervisor and product manager, built its layouts: multi-agent router, versioned AI generations with human approval, tool calls, **two real-time voice agents** (one on Gemini Live).
- Modernized an existing internal user management app (admin gateway): UI/UX, complementary features, simpler flows.
- Made long-running AI jobs reliable: asynchronous processing, persisted state, retries and LLM output validation.
- Improved **front-end performance** with TanStack Query caching, IndexedDB and pagination, profiled with Chrome DevTools.
- Instrumented products with Google Analytics, with telemetry fed into an internal observability tool.
- Prototyped apps as single-file HTML for demos, sometimes in a few hours, then rebuilt them in Next.js or Vite with AI coding agents, checked with **TDD, Vitest integration tests and Playwright E2E tests**.

Software Engineering Intern (first internship) — Sep 2025 – Dec 2025

TypeScript · React · Next.js · IndexedDB · MongoDB · n8n · Airtable · Playwright

- Built full-stack, with another intern, a **production client app** for technical drawings: two AI agents extract the data, a third compares them with its own and keeps the best value per field (**confidence score**); the app then computes material and cost.
- Built v1 of an internal project lifecycle tool with LLM agents (n8n, Airtable), **adopted internally across company products**.
- Built an AI image generation and editing app on a **node-based canvas** where users chain steps (generation, image fusion, edits).

Université de Sherbrooke — Sherbrooke, QC

Research Assistant – Spatio-temporal Forecasting — Sep 2026 – Present

Supervisor: Prof. Djemel Ziou · Python · ST-LLM+

- Forecasting emergency call volumes per county (MRC) and 8-hour shift with a spatio-temporal LLM.

Teaching Assistant – IFT320 Operating Systems — Jun 2026 – Aug 2026

- Graded C/Linux assignments for **38 students**, with feedback on synchronization, concurrency and code quality.

Research Assistant – Application Development — Feb 2026 – May 2026

Supervisor: Prof. Djemel Ziou · React · Vite · Three.js · D3.js · WebXR

- Built a curriculum explorer: **interactive 3D graph of courses**, grid views and an AI assistant restricted to controlled actions.
- Built an ETL pipeline for a CCSLL application in March 2026 (consulting contract).

Borealis — Magog, QC • Remote

Web Developer Intern — Jan 2025 – Apr 2025

JavaScript · SQL · XML · Jira

- Delivered client-requested features from Jira tickets; traced and fixed a recurring interface crash with SQL/XML patches.

PROJECTS

Nexora, AI-Augmented Learning Platform – Capstone Project (team lead) — Sep 2026 – Present

Supervisor: Prof. Nadia Tahiri • IFT596, team of 5 • my areas: full-stack, AI/RAG, UI/UX • in development

- Leading a team of 5: a RAG tutor that cites its sources, a dashboard of misunderstood concepts and a course-content agent.

Shared Infrastructure to Ship Apps Fast – Personal Project — Oct 2026 – Present

- Template repo per app, shared Supabase auth and database with **per-app isolation** (row-level security), Cloudflare Pages.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Java, C#, C/C++, Bash; bilingual communication (French and English)
Front-End: React, Next.js, Vite, Tailwind CSS, HTML/CSS, TanStack Query, Redux, IndexedDB, Three.js, D3.js, WebXR, UI/UX design
Web & Product: responsive design, accessibility, SEO, A/B tests, feature flags, CMS-style publishing workflows, Google Analytics
Back-End & Data: Node.js, REST APIs, MongoDB, PostgreSQL, Supabase, BigQuery (Google Cloud), schema design, microservices
AI & Agents: OpenAI, Claude and Gemini APIs, RAG, embeddings, pgvector, prompt engineering, tool calls, voice agents, MCP, n8n
LLM Quality & Ops: LLM evaluation, cost and token tracking, LLM call logging and tracing, LangGraph, LangChain
Testing & DevOps: TDD, Jest, Vitest, Playwright (integration, E2E), ESLint, CI/CD, GitHub Actions, Docker, Vercel, Agile/Scrum
AI-Assisted Development: Claude Code, Codex, GitHub Copilot, Cursor, Gemini CLI, skill design

EDUCATION

B.Sc. in Computer Science (Co-op), Université de Sherbrooke, Dec 2026 • AWS Certified Solutions Architect (in progress)